

Multimodal Automotive Knowledge Graph for Surface Defect Diagnosis in Engine Block Casting using Neo4j

LITERATURE SURVEY REVIEW

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1. Introduction

Background

Industrial fault diagnosis in manufacturing relies heavily on technical documentation, machine logs, and quality inspection reports. On automated casting lines, equipment operators and maintenance engineers must quickly cross-reference error symptoms with operating procedures, equipment manuals, and replacement parts. Traditional relational systems store this information in separate databases, making multi-step lookups slow during production line stoppages.

Case Study Scope

This project focuses on a specific case study: surface defect diagnosis in engine block die-casting. High-resolution inspection cameras capture defects like gas porosity, thermal cracks, blowholes, and cold shuts on the casted blocks. This project models these visual defects, machine parameters, repair guides, visual diagrams, and spare parts in a connected Neo4j knowledge graph.

2. Graph Database Performance vs. Relational Systems

Traversal Efficiency and Index-Free Adjacency

Property graph databases use index-free adjacency, where each node stores direct memory references to its connected neighbor nodes. Traversing a relationship is a constant-time operation rather than a dynamic join computed at query execution time. For diagnostic queries traversing multi-tier connections (such as Inspection Image to Defect Type to Casting Machine to Repair Procedure to Spare Part to Warehouse Bin), graph databases avoid the performance cost of multi-table SQL JOIN operations.

Performance Trade-offs

Literature comparing graph databases and relational engines shows clear trade-offs based on query workload:

- **Multi-entity Traversals:** Benchmarks by Sandell et al. (2024) comparing Neo4j, MySQL, and ArangoDB showed that Neo4j executes multi-hop pattern-matching queries significantly faster and with lower memory overhead than MySQL.

- **Global Aggregations:** Work by Kotiranta et al. (2022) indicates that while Neo4j outperforms relational engines on short to medium graph traversals, SQL engines perform better on global key-based aggregation queries.
- **Graph Engine Benchmarks:** Comparative evaluations by Monteiro et al. (2023) across graph databases showed Neo4j achieved the lowest execution time, lowest average CPU load, and lowest memory usage among evaluated systems.

3. Existing Work on Industrial Knowledge Graphs

Text-Based Extraction and Ontologies

Knowledge graphs have been increasingly applied to equipment maintenance. Chen et al. (2025) built a fault diagnosis knowledge graph for production lines using an ALBERT-BiLSTM-Attention-CRF pipeline to extract entities from textual logs into Neo4j. Tang et al. (2023) constructed a schema layer and extraction pipeline for aircraft maintenance manuals to enable diagnostic question-answering. Huang and Song (2025) demonstrated using Large Language Models to automate the extraction of diagnostic triples from text manuals into Cypher statements.

Graph-Driven Diagnostics

Research by Cai et al. (2024) combined knowledge graphs with Bayesian reasoning to isolate equipment failure root causes. Mao et al. (2024) used Graph Convolutional Neural Networks over fault graphs for machine diagnosis, while Xia et al. (2023) combined knowledge graphs with Graph Neural Networks to recommend maintenance plans. Han et al. (2022) studied the ongoing schema evolution of fault graphs as new machinery failure modes appear.

4. Literature Gap Analysis

Unimodal Text Dependency

Existing industrial diagnostic knowledge graphs in literature (Chen et al., 2025; Tang et al., 2023; Huang & Song, 2025) rely almost exclusively on text data extracted from PDF manuals and logs. However, modern casting inspection is visual. Camera systems produce inspection images with bounding-box coordinates, and maintenance staff rely on visual SOP diagrams. Current studies lack a model for storing image nodes, image metadata, and visual diagram paths directly inside property graph databases.

Overly Generic Scope

Much of the published literature provides high-level frameworks for generic 'smart manufacturing' (Wan et al., 2024) without detailing concrete failure modes. High-level abstractions do not account for specific domain parameters, such as iron/aluminium casting parameters (mold temperature, injection pressure) and specific surface defect classifications.

Disconnection from Inventory and Warehouse Data

Published diagnostic knowledge graphs usually stop at recommending a textual repair procedure. They do not link the fault to real-time inventory databases. Knowing what procedure to perform is ineffective if the required spare part is out of stock or its physical warehouse bin location cannot be retrieved in the same query.

5. Proposed System

Multimodal Node Structure in Neo4j

To address these gaps, this project builds a Multimodal Automotive Knowledge Graph in Neo4j focused on Engine Block Casting Surface Defect Diagnosis. The schema explicitly models visual data alongside structured text and inventory attributes:

- Camera inspection images and SOP visual assembly diagrams are represented as Image nodes containing image file paths, timestamps, and confidence scores.
- Bounding-box spatial annotations are stored as properties on relationships connecting camera images to defect types.
- Equipment nodes store die-casting machine IDs and line locations.

End-to-End Diagnostic and Inventory Traversal

The proposed graph connects the complete operational path:

Inspection Image -> Defect Type -> Casting Equipment -> Repair Procedure -> Visual SOP Diagram -> Spare Part -> Warehouse Bin Location.

Executing a single Cypher query allows an operator viewing a flagged inspection image to instantly retrieve the defect name, originating machine, visual repair guide, required spare part, current stock count, and warehouse bin location.

6. References

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